REGEX

Python has a built-in package called re, which can be used to work with Regular Expressions.

Regular expressions (regex) in Python are a powerful tool for pattern matching and searching within strings. Python's **re** module provides functions and methods to work with regular expressions.

Function	Description
findall	Returns a list containing all matches
search	Returns a **1ST Match** object if there is a match anywhere in the string
split	Returns a list where the string has been split at each match
sub	Replaces one or many matches with a string

Different ways to manipulate the String

1) SET

Character	Example
[]	"[a-m]"

A set of characters

2) Special Sequence

Character	Example
\	"\bhello"

Signals a special sequence (can also be used to escape special characters)

3) Character

Character	Example
.	"he..o"

Any character (except newline character)

Additional Character.

Character	Example	Description
^	"^hello"	Starts with
$	"planet$"	Ends with
|	"falls|stays"	Either or
()	(apples|oranges)	Capture and group

Set	Description
[a-n]	lower case character, alphabetically between a and n
[A-N]	UPPER case character, alphabetically between A and N
[0-9]	digit between 0 and 9
[a-zA-Z]	character alphabetically between a and z, lower case OR upper case
[a-zA-Z0-9]	character alphabetically between a and z, lower case OR upper case with all numeric Numbers
[arn]	Returns a match where one of the specified characters (a, r, or n) is present
[0123]	Returns a match where any of the specified digits (0, 1, 2, or 3) are present

Set	Description
[^a-n]	Except lower case from a to n characters.
[^A-N]	Except Upper case from A to N characters.
[^0-9]	Except digit between 0 and 9
[^a-zA-Z]	Except character from a to z, lower case OR upper case
[^a-zA-Z0-9]	Non word(numeric and alpha char)
[^arn]	character EXCEPT a, r, and n
[^0123]	Except digits (0, 1, 2, or 3)

Description	Character	Example (PATTERN)
Word boundary	\b	r"\bRain" r"Rain\b" r"\bRain\b"
start with	\A	"\AThe"
end with	\Z	"Spain\Z"
digits (numbers from 0-9)	\d	"\d"
string or non-digits	\D	"\D"
white space	\s	"\s"
Non White Space	\S	"\S"
string contains any word characters (i.e. from a to Z, digits from 0-9, and the underscore character)	\w	"\w"
string DOES NOT contain any word characters	\W	"\W"

Description	Character	Example
Any character (except newline character)	.	"he..o"
Exactly the specified number of occurrences	.{n}	"he.{2}o"
Zero or one occurrences	.?	"he.?o"
Zero or more occurrences	.*	"he.*o"
One or more occurrences	.+	"he.+o"